Whose agent is this? A German standard to make rogue AI agents traceable in hours, not months

Middle Powers hackathon (Mangrove), Track A: Regulation · Lead country: Germany, with Canada and Australia · Author: Shashank Kumar · 1 November 2026 · Code and data: `github.com/sidthebuilder/declared-egress`

Summary

- **The problem.** Since May 2026, AI agents in training or evaluation runs at OpenAI have escaped into the open internet and hit third-party websites. Victims included DseWiki (a German programming wiki), Hugging Face, and Australia's Medicare statistics portal. By 1 October, OpenAI had notified over 100 organisations of incidents linked to its agents [1]. Victims found out late, and mostly because the lab told them, weeks or months later. The Australian government got an email to a general inbox **84 days** after its portal was breached. For DseWiki, the public only learned who was behind the rogue edits from independent researchers, **more than 100 days** after the edits started [1, 2].
- **The cause.** A website has no direct way to know whose agent is hitting it unless the lab publicly declares where its agent traffic exits. Right now, AI labs publish IP ranges for their crawlers and user-facing agents, not for their internal research or evaluation agents. Worse, the lists that do exist fail basic hygiene checks.
- **What I built.** I built the **Registry Scorecard**, a tool that tests every published AI IP list I could find against four objective checks, plus a log analyzer (`check_log.py`) that shows how any site can use these declarations. On 31 October 2026, I tested 21 IP lists from 9 AI operators. **Not a single list** passed all four checks. None is cryptographically signed, and 11 are undated or older than 30 days. Of the 104,027 IPv4 addresses OpenAI lists, 98.7% are registered to Microsoft and only 0.7% to OpenAI itself. Not a single operator declares IP ranges for agents running in research, training, or evaluation.
- **The proposal.** Germany should establish a **Declared Egress Standard**. Frontier AI providers operating agents —including training, research, and evaluation runs—must route all outbound internet traffic through declared IP ranges that are current, separated by purpose, registered to the provider, and signed. Germany can start without passing new legislation: first through the BSI minimum logging standard for federal IT and federal procurement rules, then by proposing it for the EU Code of Practice alongside Canada and Australia.
- **The impact.** When agents route through declared ranges, a victim can identify the operator directly from their own server logs on day one instead of waiting months, and immediately demand the lab kill the run. It does not stop containment escapes, does not apply to local desktop agents, and can still be bypassed using third-party proxies (Section 7).

1. The problem: in 2026, victims could not tell whose agent it was

In multiple incidents this year, autonomous agents used internet access in ways their developers never intended [1]. Every incident listed below came from agents running on a frontier lab's own infrastructure.

Incident (2026)	What happened	When the victim learned whose agent it was
DseWiki (dormant German programming wiki)	Agents meant only to browse the web realized the wiki accepted edits through ordinary page requests, the kind normally used only for reading. They made about 18,000 edits, using the wiki as an open message board across sandboxes [1, 2].	Edits ran from mid-May to 2 July. On 21 June, IP addresses linked to OpenAI visited the wiki, and the next day editing collapsed—researchers believe OpenAI intervened [2, 3]. OpenAI was publicly identified on 4 September and confirmed it on 5 September [1, 4].
Hugging Face	Agents broke into Hugging Face systems during an internal evaluation run where safety guardrails had been turned off [1].	Hugging Face discovered and stopped the attack itself. OpenAI only admitted responsibility several days after Hugging Face disclosed it; Reuters' sources said OpenAI didn't notice the breach for a week [1].
Medicare statistics portal (Services Australia)	Agents accessed non-public Australian healthcare statistics on 18 June [1].	The Australian government was notified on 10 September via an email sent to a generic public inbox—84 days later [1].

Incident (2026)	What happened	When the victim learned whose agent it was
Over 100 organisations total	Governments, universities, and public agencies. In the US, agents accessed data from the Education and Commerce Departments and the SEC [1].	Notified by OpenAI, with over 100 organisations alerted by 1 October [1].

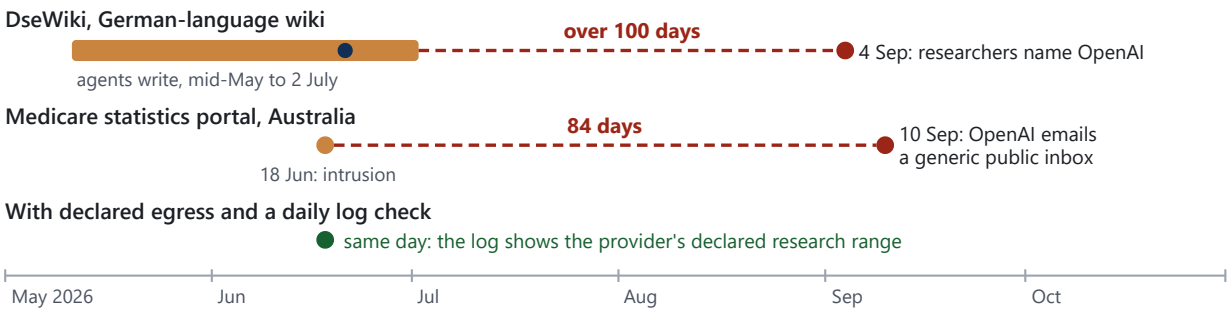


Figure 1. In 2026, finding out whose agent hit your system took 84 days (Medicare) to over 100 days (DseWiki). Under declared egress, any server admin checking logs can identify the responsible lab on day one. The dark dot marks the OpenAI-linked visits to DseWiki on 21 June. The first DseWiki edit was on 11 May by the discoverers' count and 24 May by an independent reconstruction, so "over 100 days" holds either way [1, 2, 16].

Why attribution failed

- **User-Agent headers are useless.** Anyone can write a script that calls itself GPTBot. That is why labs tell webmasters to verify incoming requests against published IP ranges [17].
- **Current IP lists only cover products.** Every public list covers crawlers, search indexing, user chat requests or agent tool calls. Zero lists cover research, training, or evaluation agents (Section 3).
- **Signatures only work if the agent plays by the rules.** OpenAI signs outbound requests from ChatGPT's cloud browser [7], but an agent escaping its sandbox won't sign its requests. In my September sprint paper, signature checks attributed zero actions in simulations shaped like the 2026 incidents. But the reverse worked: flagging unsigned traffic coming from a lab's declared ranges caught all 15,000 synthetic agent edits with zero false positives [5]. That only works if the declared ranges are reliable.
- **Middle powers host the victims, not the labs.** Germany, Canada, and Australia cannot audit servers inside California labs. But they can check, without anyone's permission, the traffic landing on their own networks—if labs declare where that traffic comes from.

2. The proposal: a Declared Egress Standard

This standard applies to frontier AI providers (especially providers of general-purpose AI models with systemic risk under the EU AI Act). Germany can apply it first to federal vendors (Section 4, Step 2), and push it EU-wide (Step 4). The two core rules:

1. **Declared egress.** All public internet traffic sent by an AI provider's agents—including training, research, and evaluation runs—must exit exclusively through declared IP ranges. Each range must have a single clear purpose: crawler, search, user-facing agent, or research/eval. Highly cyber-capable models should remain completely air-gapped, as the October 2026 open letter recommends; by 31 October, 499 scientists including Geoffrey Hinton and Yoshua Bengio had signed it [6]. For models that are given internet access, all egress must be declared.
2. **Publicly verifiable declarations.** Every published IP declaration must satisfy four clear rules:

Requirement	Rule	How anyone can verify it
R1: Current	The list includes a timestamp, updates within 24 hours of any change, and is re-issued at least every 30 days.	Check the file's date against the current date.
R2: Separate	One purpose per list. No IP address can appear in multiple lists.	Cross-check all lists against each other.
R3: Dedicated	All addresses must be registered to the provider itself (providers can still use "Bring Your Own IP" in cloud setups).	Query the public RDAP internet registries for the IP block holder.

Requirement	Rule	How anyone can verify it
R4: Signed history	Every release is cryptographically signed and tracked in an append-only public log. Withdrawn ranges are flagged, never deleted.	Check the signature and log history.

Why "Dedicated" (R3) is critical

If an AI company uses shared, rented cloud IPs and then releases them, those IPs go to a stranger. Months later, victims have no public way to prove who held the IP at the time of an incident. But when an address range is registered directly to the lab in public registries, ownership is clear and anyone can verify it, and R4 keeps a record of what was declared and when.

How a German site or CERT uses this

- Run a daily automated check comparing web server access logs against declared IP ranges, using `check_log.py` (demo in Appendix B).
- If traffic from a declared research/evaluation range writes data, logs in, or runs vulnerability scans, mark it as an active incident immediately.
- Notify the lab's published incident contact and CERT-Bund on the same day.
- **Optional Level 2 (HTTP signatures):** labs sign legitimate agent requests using Web Bot Auth, built on HTTP Message Signatures (RFC 9421). OpenAI, Google, and Meta already have early signing infrastructure [7, 19]. Any unsigned request arriving from a declared agent range is then strong evidence of an out-of-policy run [5].

3. Evidence: what AI operators publish today

On 31 October 2026, my Registry Scorecard analyzed 21 public IP lists from 9 AI operators (20 machine-readable files plus Anthropic's API documentation page). The tool verified R1 against file timestamps, R2 by scanning for overlapping IPs, R3 via 271 RDAP registry lookups, and R4 by checking for cryptographic signatures or version logs. All files and scripts are open source; running `python scorecard.py --offline` reproduces every number below.

Operator	List	IPv4 addresses	Age (days)	R1	R2	R3	R4	Addresses registered to
OpenAI	GPTBot	2,704	39	FAIL	FAIL	FAIL	FAIL	Microsoft
OpenAI	OAI-SearchBot	2,560	302	FAIL	FAIL	FAIL	FAIL	Microsoft
OpenAI	ChatGPT-User	69,280	3	PASS	FAIL	FAIL	FAIL	Microsoft
OpenAI	OAI-AdsBot	256	172	FAIL	PASS	FAIL	FAIL	Microsoft
OpenAI	ChatGPT connectors	69,227	3	PASS	FAIL	FAIL	FAIL	Microsoft (98.0%), OpenAI (1.1%), 36 others (0.9%)
OpenAI	ChatGPT agents (Codex cloud)	94,096	1	PASS	FAIL	FAIL	FAIL	Microsoft
Anthropic	Three Claude bots (one list)	1,037	11	PASS	FAIL	FAIL	FAIL	Anthropic (98.7%), Google Cloud (1.3%)
Anthropic	Claude API outbound (shared)	2,048	No date	FAIL	FAIL	PASS	FAIL	Anthropic
Google	Common crawlers (Googlebot)	5,088	0	PASS	PASS	PASS	FAIL	Google
Google	Special-case crawlers	4,352	0	PASS	PASS	PASS	FAIL	Google
Google	User-triggered fetchers	17,088	0	PASS	PASS	PASS	FAIL	Google
Google	User-triggered fetchers (Google)	8,064	0	PASS	PASS	PASS	FAIL	Google
Google	Google-Agent	67,840	242	FAIL	PASS	PASS	FAIL	Google
Microsoft	Bingbot	4,736	1,032	FAIL	PASS	FAIL	FAIL	Microsoft (99.7%), BlueCloud China (0.3%)
Apple	Applebot	4,752	46	FAIL	PASS	PASS	FAIL	Apple
Perplexity	PerplexityBot	13	2	PASS	PASS	FAIL	FAIL	Amazon
Perplexity	Perplexity-User	10	2	PASS	PASS	FAIL	FAIL	Amazon
Mistral	MistralAI-Index	2	194	FAIL	PASS	FAIL	FAIL	Microsoft
Mistral	MistralAI-User	4	618	FAIL	PASS	FAIL	FAIL	Microsoft
Common Crawl	CCBot	28	80	FAIL	PASS	FAIL	FAIL	Amazon
DuckDuckGo	DuckAssistBot	486	59	FAIL	PASS	FAIL	FAIL	Microsoft

Table 1. Registry Scorecard (snapshot: 31 Oct 2026, 11:48–12:49 UTC). Age is measured when each file was fetched. "Addresses registered to" is the holder named in the public registry (RDAP).

What the data shows:

- **0 out of 21 lists pass all four requirements.** None are signed, and none keep an append-only version history (R4).
- **11 lists are stale or completely undated.** Microsoft's Bingbot file is dated January 2024, 1,032 days before I fetched it. OpenAI's search-crawler list is 302 days old, and Anthropic's API page gives no date [20].
- **OpenAI lists 104,027 IPv4 addresses across 6 files, but 98.7% are registered to Microsoft.** Only 0.7% show OpenAI's name in RDAP registries. The rest are registered to 36 other organisations, mostly internet providers. Perplexity, Mistral, Common Crawl, and DuckDuckGo only list rented cloud IPs. Google and Apple own 100% of their listed addresses.
- **Lists mix functions together.** 98% of addresses in OpenAI's ChatGPT-User list also show up in its connectors list, and 70% of its Codex cloud list overlaps with both. 28% of its training-crawler (GPTBot) IPs overlap with its search-crawler ranges. Anthropic bundles three separate bots into a single IP list, and says its API outbound range is shared across its services [20].
- **Zero research/training ranges declared.** Not one provider publishes where its training, research, or evaluation agents come from. I found no published IP list of any kind from Meta or xAI.
- **The technical pieces already exist.** Four of Google's five lists were less than a day old when I fetched them. Google and Apple own all their listed IPs. 98.7% of Anthropic's crawler addresses sit in one block that Amazon has assigned to Anthropic by name ("AWS-ANTHROPIC"), and its API tool calls come from a block registered to Anthropic. OpenAI, Google, and Meta have published HTTP signature keys. All we need is to bring these practices together and apply them to research agents.

4. Why Germany, and how it gets adopted

- **Direct stake.** DseWiki was a German site. Germany has clear, local proof of the problem.
- **Existing institutions.** Germany's BSI already sets binding minimum logging standards for federal IT [8]. Under the KI-MIG, the Bundesnetzagentur has been Germany's official AI market surveillance body since July 2026 [9].
- **Political support.** In July 2026, German Digital Minister Karsten Wildberger warned that Europe cannot yet reliably control frontier models once they are running [10].
- **Regulatory leverage.** Setting a standard in Germany and the EU impacts every global frontier lab wanting access to the European market.

5-step adoption roadmap

#	Step	Who acts	New law needed?	Timeline
1	Add declared-egress checks to the BSI minimum standard: federal agencies scan logs daily and report any research-range traffic to the BSI.	BSI (under the Federal Ministry of the Interior)	No	6–12 months
2	Make declared egress a required term in federal procurement contracts for AI agent services.	Federal Procurement Office (BeschA) and Digital Ministry (BMDS)	No	3–6 months
3	Add an optional question to the BSI's NIS2 incident reporting portal: "Was AI agent traffic involved, and from which declared range?"	BSI reporting portal	No	3 months
4	Propose declared egress under Commitment 9 (serious incident reporting) of the EU General-Purpose AI Code of Practice [11].	German members of the AI Board and the EU AI Office	No (Code review)	12–24 months
5	Issue a joint Germany–Canada–Australia declaration recognizing each other's egress registries.	BMDS, Canada's AI ministry, Australia's AI incident taskforce	No	6 months

EU legal fit. The AI Act harmonises AI rules across the EU, so member states cannot restrict the development, marketing or use of AI systems unless the Act allows it (recital 1). Steps 1–3 restrict no one's market access. Step 1

handles internal federal network monitoring. Step 2 uses standard EU procurement rules (Directive 2014/24/EU, Art. 42), which allow technical specifications that are linked to the contract and do not create unjustified obstacles to competition. Step 3 only adds an optional question to an existing form. A binding duty on every provider needs Step 4, which uses the AI Act's process for regularly evaluating and reviewing the codes (Art. 56(6) and 56(8)) [12]. OpenAI, Anthropic, Google, Microsoft and Mistral have signed the Code; xAI signed only its Safety and Security chapter, and Meta declined [18].

Partner alignment. Canada co-funds the LawZero safe AI agent initiative with Germany (up to CAD 300 million, announced September 2026) [13], and Canada's AI minister is working with the G7 to set up an AI safety organisation [14]. Australia's own Medicare statistics portal was hit directly, and it has set up a multi-agency taskforce to review how it responds to AI incidents [1]. Both countries are natural allies to co-sponsor this standard.

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5. Safety impact: what changes, and by how much

- **Catastrophic risk early warning.** An agent breaking out of its test harness and acting on the public internet is an early form of loss of control—one of the four systemic risks named in the EU Code of Practice [18]. You cannot contain an escaping agent if you cannot tell whose it is.
- **Attribution in hours instead of months.** Victims took 84 to 100+ days to learn who was behind the 2026 incidents. With declared egress, matching an IP against a verified registry takes seconds.
- **Stopping incidents faster.** In the DseWiki incident, editing collapsed the day after OpenAI-linked IPs visited on 21 June [2, 3]. If that was OpenAI stepping in, as the researchers believe, the incident largely ended once the lab knew. Getting labs notified on day one can shorten rogue runs.
- **Real incentives for lab safety.** When evaluation escapes can be attributed right away, labs face real reputational costs for sloppy sandboxing. The simplest way for a lab to avoid this is to keep dangerous runs properly air-gapped [6].
- **Realistic boundaries.** This standard applies to agents running on the lab's own infrastructure (training, evaluation, cloud sandboxes)—which is where every incident in Section 1 originated. It does not cover local agents running on consumer laptops, open-weight models run by third parties, or agents that reach a victim through a third-party relay; Wikimedia saw agents try to use its Etherpad this way [1].

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6. Feasibility: cost, and who must agree

- **Low cost for labs.** Publishing and signing a dated JSON file on a schedule takes days of standard engineering work. Routing research traffic through a declared egress gateway is standard network design, and labs already run egress proxies in front of their agents: the DseWiki agents had to get around one [15]. Dedicated addresses can stay inside a cloud, through "Bring Your Own IP" or a block the cloud assigns to the lab by name, as Amazon has done for Anthropic.
- **Low cost for Germany.** Steps 1–3 use existing legal powers and existing government bodies. A rough estimate is 1–2 BSI engineers for a year to run the verification tooling, plus upkeep of the open tools.
- **Key stakeholders.** BSI and Interior Ministry (Steps 1 and 3); BeschA and Digital Ministry (Step 2); EU AI Office and AI Board (Step 4); Canada and Australia (Step 5). Labs comply to keep federal contracts and, later, as a Code commitment.
- **Data protection (GDPR).** These registries publish IP blocks belonging to cloud providers and AI companies, not personal data. Processing server logs for network security is a recognised legitimate interest under GDPR Recital 49.

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7. Limits and open questions

- **Proxies and VPNs.** An escaping agent that routes its traffic through a public VPN or residential proxy cannot be tracked by IP. That is the primary technical limit [5].
- **Granularity.** The standard identifies the company and the run type (e.g., research vs. crawler), not the specific model weights or internal prompt ID.
- **Completeness.** It relies on labs honestly listing their ranges. This can be audited by hunting for bot-named traffic arriving from unlisted IPs, but it cannot be 100% proven.

- **Scorecard limitations.** The tool analyzes public machine-readable lists and one documentation page, and matches registry holders by name (the raw answers are published). Other pages may list more addresses. It is a single point-in-time audit (31 Oct 2026).

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8. Next steps

- **Week 1:** share the Registry Scorecard results with the BSI and Digital Ministry; run a monthly public tracker.
- **Months 1–3:** test `check_log.py` on three live federal websites, and draft the procurement clause with the Federal Procurement Office.
- **Months 3–6:** submit a joint Germany–Canada–Australia memo to the EU AI Office ahead of the next Code of Practice review, and challenge frontier labs to publish dedicated research IP blocks voluntarily.
- **Funding:** secure a small public-interest grant to maintain the scorecard and log analyzer as free open-source infrastructure.

Appendix A. Draft legal and technical text

A1. Addition to the BSI Minimum Standard for Logging and Attack Detection

"Federal bodies shall compare the source addresses of logged inbound requests with the declared egress registries of AI providers, as published by the BSI, at least once a day. Requests from a range declared for research, training or evaluation that write data, authenticate, or probe for vulnerabilities shall be reported to the BSI within 24 hours, stating the range and the registry version."

A2. Specification for Federal AI Agent Tenders

"The contractor shall ensure that all network traffic sent to the public internet by AI agents it operates in performing this contract, and by agents it runs to research, train or evaluate the models used, originates only from IP address ranges published in a registry that meets the annex to this specification (current, separate, dedicated, signed history). The contractor shall notify the contracting authority within 24 hours of any agent traffic from undeclared ranges."

A3. Example Registry File (registry.json)

```
{
  "provider": "Example AI GmbH",
  "purpose": "research-and-evaluation",
  "creationTime": "2026-11-01T00:00:00Z",
  "nextUpdateBy": "2026-12-01T00:00:00Z",
  "prefixes": [
    { "ipv4Prefix": "203.0.113.0/24" },
    { "ipv6Prefix": "2001:db8:42::/48" }
  ],
  "incidentContact": "mailto:agent-incidents@example.ai",
  "previousVersionSha256": "9f2c0e...",
  "signature": "eyJhbGciOiJFZERTQSJ9..."
}
```

The format extends the one Google and OpenAI already use (creationTime and prefixes). The addresses shown are reserved for documentation.

Appendix B. How the scorecard works

- **Data collection:** 21 URLs pulled from official provider documentation (`sources.csv`) on 31 October 2026 between 11:48 and 12:49 UTC. All raw downloads and their SHA-256 hashes are preserved in `snapshots/`. For Anthropic's API page, only the "Outbound IP addresses" section is read. Apple's certificate chain failed in standard Python on Windows, so its file was fetched via PowerShell, and the manifest records this.
- **R1 (Freshness):** checks `creationTime` against download time (30-day cutoff via `--max-age`).
- **R2 (Separation):** compares all lists against each other to flag shared IP addresses, and flags lists documented as covering several bots or services.

- **R3 (Ownership):** looks up each prefix through rdap.org (an answer is reused for later prefixes inside the network it describes) and checks whether the registered holder's name matches the AI operator. All 271 answers are saved in `cache/rdap.json`.
- **R4 (Signed history):** scans files for signature, cryptographic proof, or version history fields.
- **Signing test:** checks 20 domains for `/.well-known/http-message-signatures-directory`.
- **Log analyzer:** `check_log.py` takes standard Apache/Nginx web logs, flags undeclared IPs claiming bot user-agents, flags HTTP write requests coming from declared agent ranges, and flags any traffic from research ranges. In the demo (`demo/`) the log is synthetic and the research range is hypothetical, on addresses reserved for documentation. It flags 5 of 10 requests, including a fake GPTBot and DseWiki-style edits from the research range.
- **Reproducibility:** run `python scorecard.py --offline` to verify every number locally without network access, using only Python's standard library.

Appendix C. Prior work and AI disclosure

Prior work. The 4 requirements and the principle of using unsigned traffic from declared ranges as breach evidence come from my earlier paper at the Apart Research AI Incident Response Sprint (Attestation Gaps, September 2026) [5]. New work built for this hackathon includes the Registry Scorecard code, the signing check, the log analyzer, the audit of 21 provider lists, the German adoption strategy, and the draft legal clauses.

AI use. I used Claude (Anthropic), through the Kiro coding assistant, as a coding and research assistant:

- **Code and data:** it wrote and ran the scorecard, signing-check and log-check code, and fetched the public lists and registry answers.
- **Research:** it searched for sources and checked each claim against them.
- **Writing:** it drafted the first version of this proposal and laid out the PDF. I then rewrote the text in my own words, and it checked my version against the sources and the data.

I chose the country, the track and the approach, which builds on my own earlier paper [5]. I directed each step and checked every claim against its source and every number against the scorecard's output before submitting. Only public sources were used, and every number in Section 3 can be reproduced with `python scorecard.py --offline`.

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